Guanyu Liu

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Research Interests: Reliable and Efficient AI; LLM Reasoning and Post-Training; Self-Evolving Agents

EDUCATION

University of Macau Macau, China
B.S. Candidate in Statistics and Data Science GPA: 3.93/4.0 *2023.08 - 2027.06*
Relevant Coursework: *Data Structures and Object-Oriented Programming, Linear Algebra, Probability Theory, Discrete Structures, Machine Learning, Pattern Recognition*

Honors & Scholarships

- IET Prize 2025 — Sole Recipient at University of Macau
- Hong Tai Day Scholarship (2024–2025) — Sole Recipient across 10 Universities
- Bank of China Scholarship (2023–2024) — One of Two Recipients at University of Macau

RESEARCH

MetaGate: Reliable Adaptive Test-Time Compute (*Accepted at PRICAI 2026*) Macau, China
Independent Researcher and Sole Author *2026.03-2026.04*

- Benchmarked Qwen3 models from 1.7B to 30B on MATH-500 and identified that uniform compute allocation outperformed a common difficulty signal in 13 of 20 cross-model transfers.
- Developed a target-label-free gate that standardizes ANLL and four-sample answer-consistency signals using frozen source statistics; their discrepancy determines whether to use adaptive allocation or fall back to uniform sampling.
- Reduced below-uniform transfers from 13 of 20 to 1 of 20 at 6.99 decoded samples per problem, averaging 0.28 percentage points above uniform and 0.79 points above a sample-budget-matched random fallback.

Dual-Replay: Evidence Dependence and Recovery in Multimodal Agents Macau, China
Research Assistant (Advisor: Prof. Long Chen, University of Macau) *2026.04 - 2026.09*

- Led method design, experiments, and writing for dual replay, crossing clean/corrupted specialist outputs with controlled/adaptive continuations to distinguish evidence dependence from recovery.
- Implemented pre-crop probes and visual-content controls; dual replay identified 48 recoveries among 917 eligible main-study system–units across four chest-X-ray agents, all missed by terminal observation deletion.
- Used calibrated compensation gains for offline evidence allocation under source corruption: +2.5 percentage points over uniform across 480 requests at 25% continuation authorization (95% CI: 1.0–4.2).

CARE: LLM-Guided Reaction Optimization Montreal, QC
Research Assistant and Co-First Author (Advisor: Dr. Tianyu Shi of McGill University) *2026.05 - 2026.08*

- Co-developed LLM-generated executable scoring rules for budgeted reaction optimization; designed a reference-conditioned controller separating program synthesis from experiment selection.
- Recorded intervention decisions and supporting evidence before outcome reveal, then updated persistent controller state to retain, revise, or regenerate programs; validated restricted execution, finite scoring, and candidate-order invariance.
- Combined budgeted exploitation/exploration with deterministic numerical fallback, ranking first across eight tasks with 30 matched seeds each in normalized regret, best-so-far AUC, and Top-1% Success@15.

IPU: Label-Free Prompt Selection Hangzhou, China
Visiting Student, MAPLE Lab (Advisor: Prof. Guojun Qi of Westlake University) *2026.06 - 2026.08*

- Led problem formulation, theory, implementation, and evaluation of IPU-PromptWeighted, a source-calibrated answer-class posterior for prompt selection under distribution shift without target labels.
- Used source-only, item-grouped nested cross-fitting with multinomial proper loss; on 40 GSM8K prompts, achieved Spearman $\rho = 0.7208$ versus Mean Pairwise Agreement (0.5713), majority pseudo-labeling (0.5583), and direct source accuracy (0.4721).
- Retained $\rho = 0.6802$ with 80% fewer target calls using Equal-z fusion of IPU and agreement; HotPotQA validation is ongoing for an ICLR 2027 submission.

RESEARCH (CONTINUED)

Lean2Isabelle: Cross-Assistant Proof Translation Macau, China
Research Assistant (Advisor: Prof. Derek Wong of University of Macau) *2025.04 - 2026.05*

- Co-first author on Lean-to-Isabelle proof translation, contributing from idea formulation through system design, evaluation, and manuscript preparation.
- Developed the entire data pipeline with Herald-ISA (26,104 verified pairs), LeanWorkbook-ISA (9,683 RL instances), and MiniF2F-DSP (217 transfer cases).
- Implemented and fine-tuned Statement-SFT and Theory-SFT by introducing GRPO for theory generation, Minimal Valid Progress shaping, a difficulty-ordered curriculum, and Isabelle/PISA verifier rewards, achieving correct statement translation on 86.0% and 85.7% of two benchmarks and improved semantic proof success from 49.8% to 54.3%.

Target-Label-Free Mitochondrial Segmentation Macau, China
Research Assistant (Advisor: Prof. Long Chen of University of Macau) *2025.05 - 2026.08*

- Developed target-label-free domain adaptation by transferring target appearance to labeled source stacks while preserving morphology, without target pseudo-labels.
- Designed a Semantic Readiness Controller to activate boundary-safe replay by assessing EMA representation drift, target image-only loss, and source-loss gains alongside MAD-estimated noise.
- Implemented a full-stack posterior-stability gate, achieving 90.85% F1 and 83.29% IoU on VNC III-to-Lucchi2 (0.14 and 0.23 points above fixed timing), with replay at iteration 2k versus 4k. Preparing an ICASSP 2027 submission.

PUBLICATIONS

Peer-Reviewed and Accepted Publications

[1] **Liu, G.** (2026). "When Difficulty Signals Are Unreliable: Auditing and Gating Sample-Count Adaptive Test-Time Compute." *PRICAI 2026*. Accepted regular paper; sole author.

[2] **Liu, G.**[*], Ma, J.[*], & Wong, D. F. (2026). "Lean2Isabelle: Factorized Cross-Assistant Proof Translation." *The 3rd AI for Math Workshop at ICML 2026*. Workshop paper; extended version under review at NeurIPS 2026. [*]Equal contribution.

Manuscripts Under Review

[3] **Liu, G.**, Wang, Z., & Chen, L. (2026). "Ignored or Recovered? Dual-Replay Provenance for Chest X-Ray Agents." Submitted to *ICASSP 2027*; under review.

[4] **Liu, G.**[*], Kong, W.[*], Tang, C.[*], Wang, Z., Zhang, B., Li, B., Zhang, P., & Shi, T. (2026). "CARE: Context-Aware Ranking Evolution with Executable Scoring Programs for Budgeted Reaction Optimization." Under review at *AAAI 2027*. [*]Equal contribution.

[5] Han, Y., Liu, B., Zhou, Z., **Liu, G.**, Zhang, Z., Yang, Y., Wang, W., Shi, I. N., Yunyan, Y., He, L., & Shi, T. (2026). "MAPGD: Multi-Agent Prompt Gradient Descent for Collaborative Prompt Optimization." Under review at *Transactions on Machine Learning Research (TMLR)*; preliminary version presented at the *NeurIPS 2025 Workshop on Scaling Environments for Agents (SEA)*.

TECHNICAL SKILLS

Programming: Python, C++, Bash
LLM Training & Post-Training: PyTorch, Hugging Face Transformers, TRL
LLM Programming & APIs: DSPy, LiteLLM, OpenAI-compatible APIs
Inference & Infrastructure: vLLM, Slurm, Git, Linux, LaTeX